

FINANCIAL MODELING & EQUITY RESEARCH CASE STUDY

Apex Solutions Inc. | Technology Sector

SELF-DIRECTED PROJECT | 5-YEAR FORECAST | FULLY LINKED 3-STATEMENT MODEL

Executive Summary

Built an integrated financial model for Apex Solutions Inc., a technology-sector case-study company with Software, Hardware and IT Services segments. The model forecasts the Income Statement, Balance Sheet and Cash Flow Statement through 2028E, with supporting schedules for working capital, CapEx & depreciation, and debt & interest. The analysis evaluates growth, profitability, liquidity, leverage and cash generation.

Revenue CAGR 18.6% 2023A–2028E	EBITDA CAGR 18.8% 2023A–2028E	Net Income CAGR 22.3% 2023A–2028E	2028E Current Ratio 5.94x Liquidity
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Company Profile	Model Scope
The supplied case-study profile describes Apex Solutions as founded in 2011, operating across Software, Hardware and IT Services, with operations in 30+ countries and 5,000+ employees.	Historical financials → segment drivers → operating costs → working capital → CapEx/depreciation → debt/interest → 3 statements → ratio analysis → financial insights.

Professional Positioning

A self-directed financial modeling / equity research case study demonstrating practical Excel skills in forecasting, statement integration, supporting schedules and financial analysis. Apex Solutions Inc. is the company analyzed in the case-study materials; it is not represented as an employment client.

Core skills: Financial Modeling • 3-Statement Modeling • Forecasting • Ratio Analysis • Working Capital • CapEx & Depreciation • Debt Schedule • Excel

1. Company & Business Analysis

Segment	Business focus	Forecast assumption
Software	Enterprise software, cloud solutions, customized development	20% annual growth
Hardware	Servers, networking, computing devices and peripherals	18% annual growth
IT Services	Cloud, consulting, managed IT and cybersecurity	16% annual growth

Operating assumptions

Driver	Forecast assumption
Software cost of sales	50% of Software revenue
Hardware cost of sales	48% of Hardware revenue
IT Services cost of sales	50% of IT Services revenue
SG&A;	15% of total revenue
Effective tax rate	23.4%
Receivables collection	19 days
Inventory processing	70 days
Supplier payment period	48 days

Analytical framing

The model is driver-based rather than a simple top-line extrapolation. Segment revenue assumptions feed cost-of-sales assumptions; SG&A; is modeled as a percentage of revenue; tax, depreciation, working capital, CapEx and debt schedules then flow through the linked statements.

Equity-research lens: growth is supported by explicit segment drivers, while stable cost assumptions preserve operating discipline.

2. Integrated 3-Statement Financial Model

Revenue Drivers Software • Hardware • IT Services	Income Statement Revenue → EBITDA → EBIT → EBT → Net Income	Balance Sheet Cash • AR • Inventory • PPE • Debt • Equity
Supporting Schedules Working Capital • CapEx & Depreciation • Debt & Interest	Cash Flow Statement Operating • Investing • Financing	Validation Balance Sheet check = 0 across forecast periods

Modeling work completed

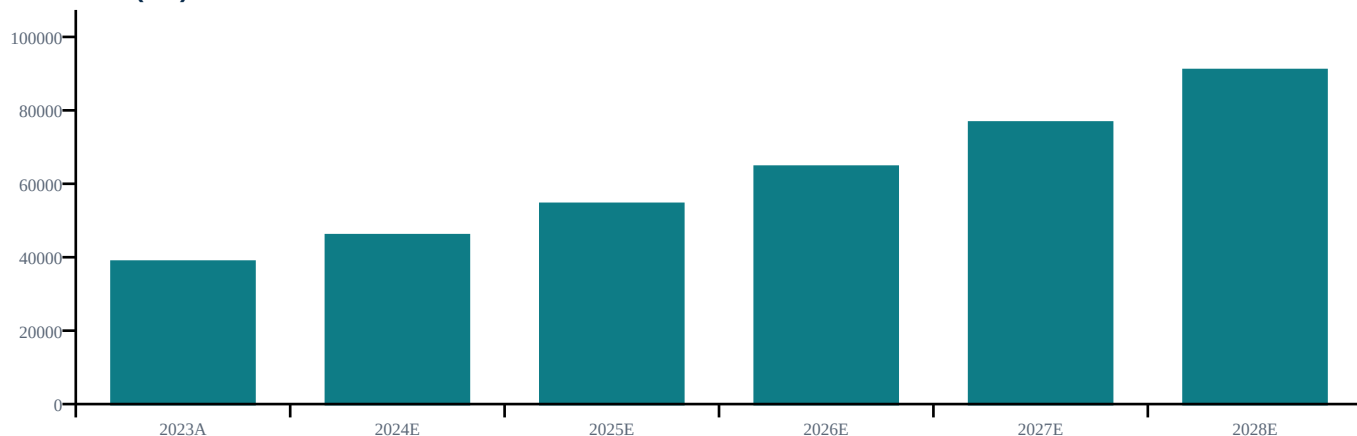
- Forecasted the Income Statement for five forecast years using segment-level revenue and cost drivers.
- Built a depreciation and CapEx schedule linking investment spending to net PP&E; and depreciation.
- Built working-capital schedules using receivables, inventory and payables days.
- Built a debt schedule with annual issuance, repayment/amortization and interest expense.
- Linked interest income to forecast cash balances and the Income Statement.
- Linked the Income Statement, Cash Flow Statement and Balance Sheet and validated the balance-sheet check.
- Calculated historical and forecast ratios to evaluate liquidity, leverage and returns.

2028E Operating CF \$24,590m Cash generated	2028E CapEx \$15,000m Investment	2028E Ending Cash \$21,741m Liquidity	2028E Debt \$20,200m Term loan
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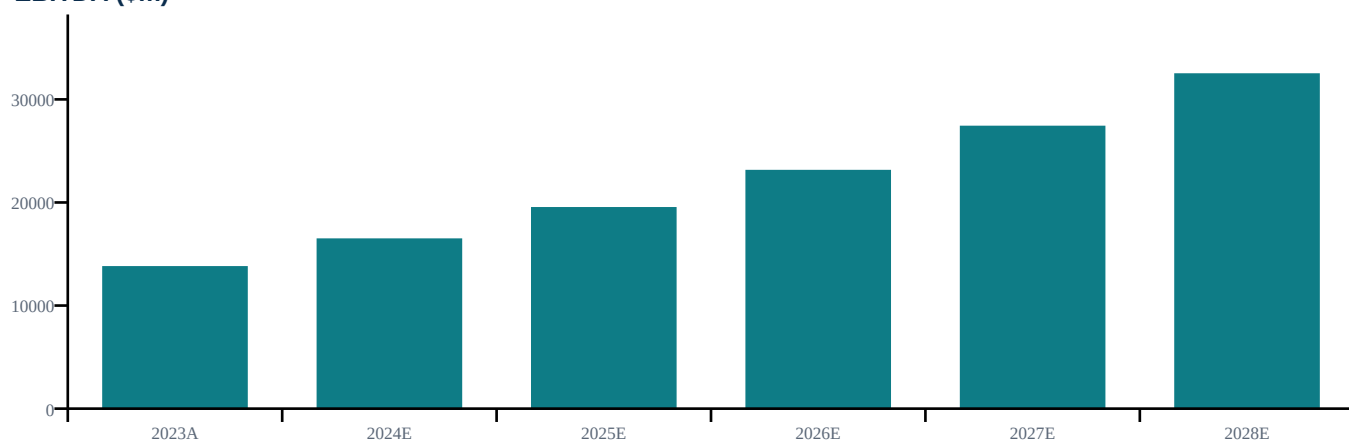
Positive operating cash flow supports continued investment, while scheduled debt repayment and rising cash materially strengthen the forecast balance sheet.

3. Forecast Financial Performance

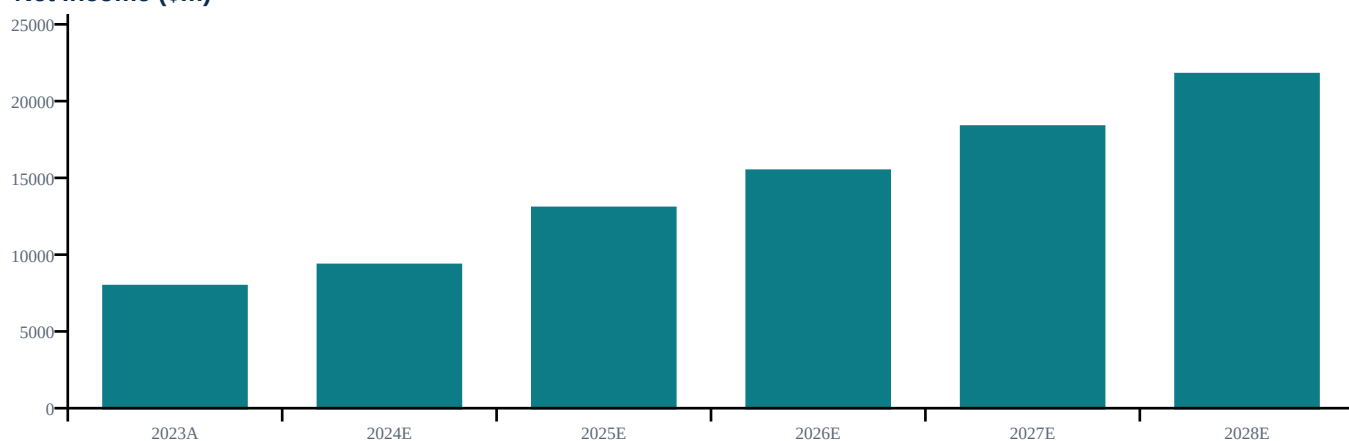
Revenue (\$m)



EBITDA (\$m)



Net Income (\$m)



Metric	2023A	2024E	2025E	2026E	2027E	2028E
Revenue	\$38,800m	\$45,994m	\$54,531m	\$64,664m	\$76,694m	\$90,976m
EBITDA	\$13,700m	\$16,388m	\$19,428m	\$23,037m	\$27,320m	\$32,404m
Net Income	\$7,950m	\$9,330m	\$13,044m	\$15,467m	\$18,342m	\$21,756m
Gross Margin	50.8%	50.6%	50.6%	50.6%	50.6%	50.6%
EBITDA Margin	35.3%	35.6%	35.6%	35.6%	35.6%	35.6%
Net Margin	20.5%	20.3%	23.9%	23.9%	23.9%	23.9%

Key observation: Revenue rises from \$38,800m in 2023A to \$90,976m in 2028E (18.6% CAGR). EBITDA grows at 18.8% CAGR and net income at 22.3% CAGR. Gross and EBITDA margins remain broadly stable, while net margin expands from

20.5% to 23.9%.

4. Ratio & Financial Health Analysis

Ratio	2023A	2024E	2025E	2026E	2027E	2028E
Current Ratio	2.67x	2.66x	3.23x	3.91x	4.90x	5.94x
Quick Ratio	1.22x	1.20x	1.77x	2.45x	3.44x	4.48x
Cash Ratio	0.51x	0.40x	0.97x	1.65x	2.64x	3.68x
Debt / Equity	1.91x	1.07x	0.65x	0.43x	0.31x	0.22x
ROE	60.2%	41.4%	36.7%	30.3%	26.4%	23.9%

Liquidity	Leverage	Returns
Current ratio rises from 2.67x to 5.94x. Quick and cash ratios also strengthen materially.	Debt/equity falls from 1.91x to 0.22x, indicating a much less leveraged balance sheet.	ROE normalizes from 60.2% to 23.9% as the equity base expands rapidly through retained earnings.

Analyst takeaway

The forecast indicates a strengthening liquidity profile and significantly lower financial leverage. The investment case is better supported by growth, cash generation and deleveraging than by an improving ROE; the decline in ROE should be interpreted alongside the rapid expansion of shareholders' equity.

5. Working Capital, CapEx & Debt Analysis

Working Capital

Metric	2023A	2024E	2025E	2026E	2027E	2028E
Receivables days	18	19	19	19	19	19
Inventory days	76	70	70	70	70	70
Payables days	53	48	48	48	48	48
Cash conversion cycle	42	41	41	41	41	41

The forecast targets 19-day receivables, 70-day inventory and 48-day payables, producing a 41-day cash conversion cycle. These assumptions drive the AR, inventory and AP balances used in the linked Balance Sheet and Cash Flow Statement.

Debt & Cash

Metric	2023A	2024E	2025E	2026E	2027E	2028E
Ending term loan	\$25,200m	\$24,200m	\$23,200m	\$22,200m	\$21,200m	\$20,200m
Interest expense	\$1,800m	\$2,223m	\$2,133m	\$2,043m	\$1,953m	\$1,863m
Interest income	\$200m	\$39m	\$69m	\$155m	\$301m	\$523m
Ending cash	\$1,400m	\$1,191m	\$3,436m	\$6,919m	\$13,151m	\$21,741m

Term debt declines from \$25,200m in 2023A to \$20,200m in 2028E, while ending cash rises from \$1,400m to \$21,741m. Interest expense falls from \$1,800m to \$1,863m.

CapEx & Depreciation

CapEx is forecast at \$10.0bn, \$11.5bn, \$13.0bn, \$13.5bn and \$15.0bn across 2024E–2028E, while depreciation is modeled at 4.4% of revenue. The schedule links CapEx and depreciation to PP&E; and then into the three statements.

6. Equity Research Conclusion

INVESTMENT VIEW — BASED ON THE MODEL ASSUMPTIONS

Apex Solutions Inc. presents a constructive financial profile under the forecast assumptions. Revenue compounds at 18.6%, EBITDA at 18.8%, and net income at 22.3% through 2028E. Liquidity strengthens materially, while debt/equity falls to 0.22x. Strong operating cash generation supports substantial CapEx while still allowing cash accumulation.

Theme	Analyst interpretation
Growth	Diversified segment growth supports a strong top-line trajectory through 2028E.
Profitability	Gross and EBITDA margins remain broadly stable; net margin improves over the forecast.
Liquidity	Current, quick and cash ratios strengthen significantly as cash balances build.
Leverage	Debt/equity falls sharply, indicating a materially less leveraged balance sheet.
Cash generation	Operating cash flow rises substantially and funds ongoing CapEx requirements.
Returns	ROE declines as the equity base expands; this is an important nuance rather than a deterioration in absolute earnings.

How to position this project

Use this case study to demonstrate practical capability in Excel-based financial modeling, financial statement forecasting, ratio analysis and investment-oriented interpretation. Be prepared to explain model architecture, assumptions, circular-reference risks, working-capital mechanics, debt/interest linkage and the Balance Sheet check.

Portfolio note: This document presents a self-directed case study. It should not be presented as client work or as an official research report issued by Apex Solutions Inc.